Unit 8, Lesson 3: Decomposing Quadrilaterals to Find Area

Benchmark

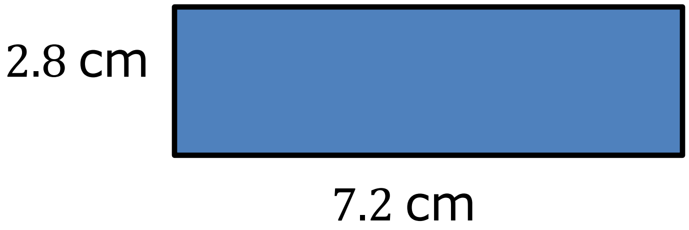
MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

Learning Target

- ☐ I can find the area of quadrilaterals by decomposing them into triangles and rectangles.

8.3.1 Warm-Up: Perimeter and Area of a Rectangle

1. A rectangle is shown with side lengths given in centimeters (cm).



a. Find the perimeter of the rectangle. Include units.

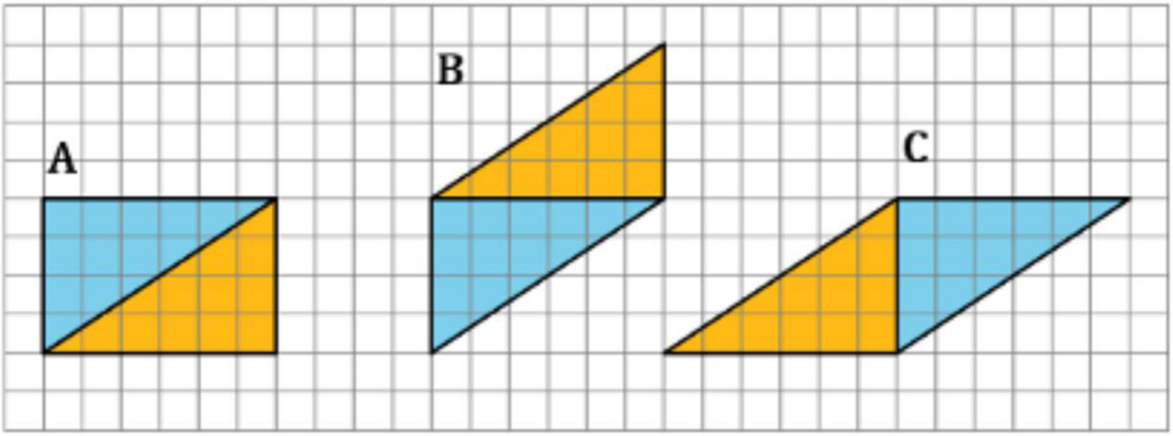
Perimeter = __________

b. Find the area of the rectangle. Include units.

Area = ___________

8.3.2 Collaboration: Decomposing Parallelograms

Three parallelograms are shown on the grid. Each square is 1 centimeter (cm) by 1 cm.



1. Work with your partner to complete the table using the figures shown in the grid. Include units.

Parallelogram	Area of Blue Triangle	Area of Gold Triangle	Sum of the Two Triangles
A			
B			
C			

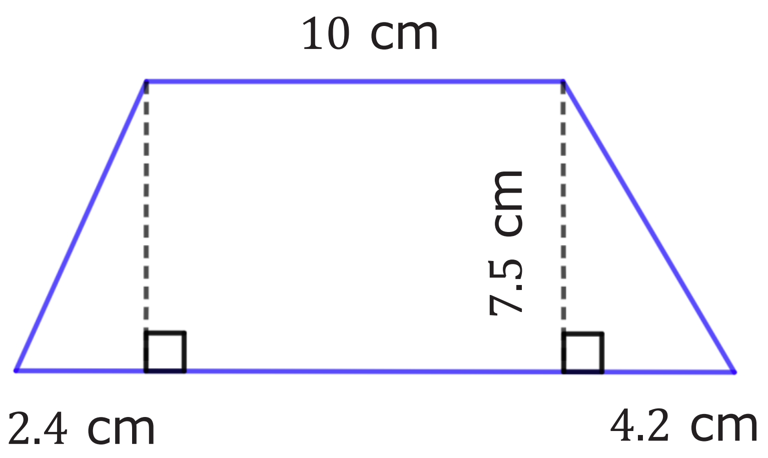
2. Discuss with your partner how the areas of the triangles are related to the area of the parallelogram they form.

   Write a brief summary of your discussion.

8.3.3 Exploration: Finding Area by Decomposing

1. A quadrilateral with measurements in centimeters (cm) is shown.

Image not drawn to scale



a. Copy the image on a piece of paper.

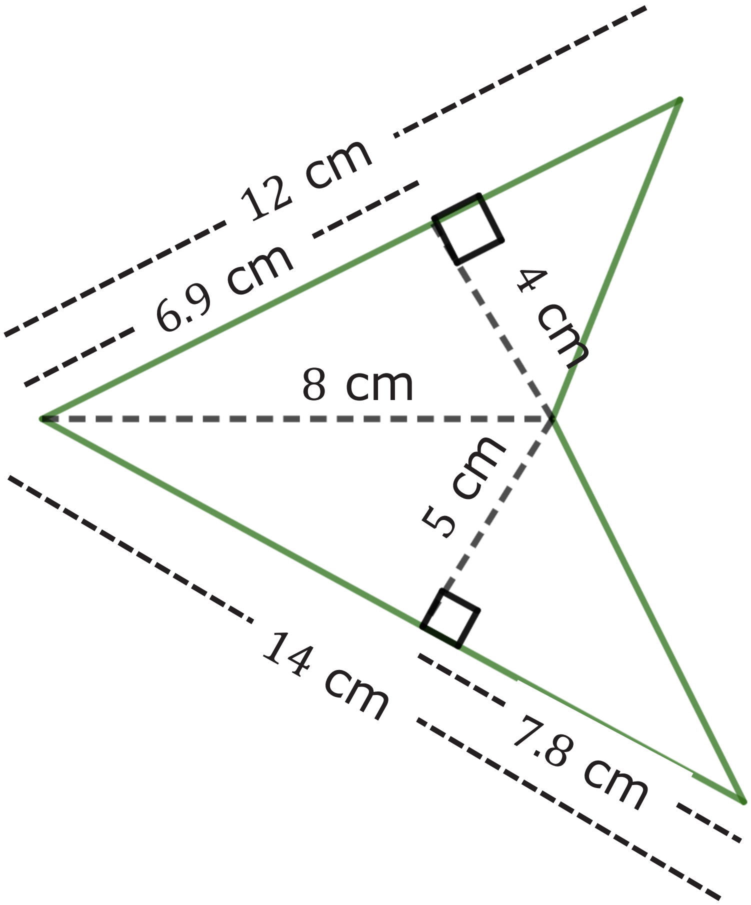
b. Decompose the quadrilateral into two triangles and a rectangle by cutting along the dotted lines. Glue or tape the three figures here.

c. Using the measurements given in the original image, label the base and height of each figure.

d. What is the area of each figure? Include units.

e. Find the area of the original quadrilateral by finding the sum of the three figures. Include units.

2. Another quadrilateral is shown.



a. Decompose the quadrilateral into triangles. Sketch the decomposed triangles here.

b. Using the measurements given in the original image, label the base and height of each triangle.

c. Find the area of the original quadrilateral by finding the sum of the areas of the decomposed triangles. Include units.

d. Sketch any other ways you could have decomposed the original figure.

To find the area of a quadrilateral, decompose the figure into rectangles or triangles. Then, find the sum of the areas of the smaller figures and add them together for the total area.

8.3.4 Guided Practice: Finding Area by Decomposing

1. Decompose the quadrilateral to find the area. Lengths are shown in feet (ft). Include units.

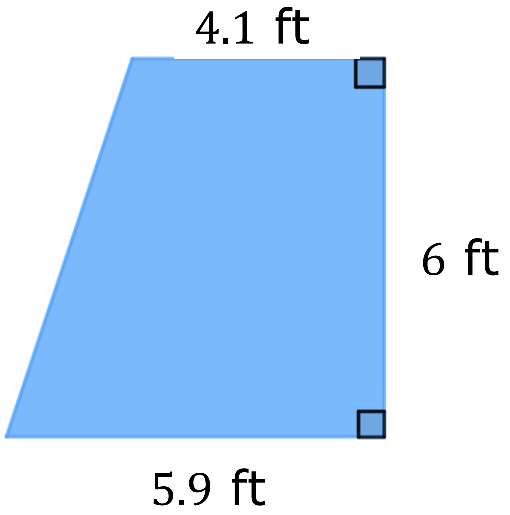
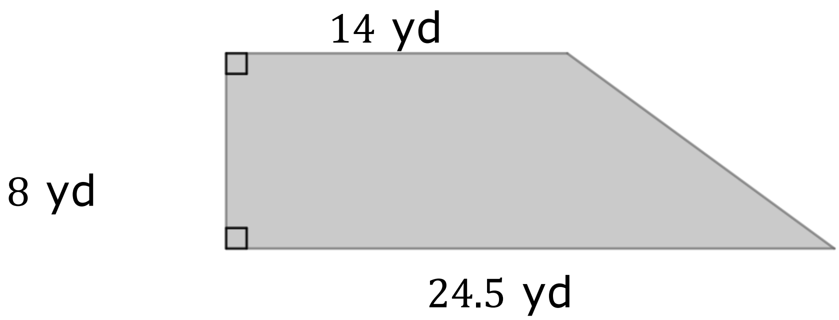


Image not drawn to scale

Area = ____________________

2. The floor plan of a store is shown with measurements in yards (yd).



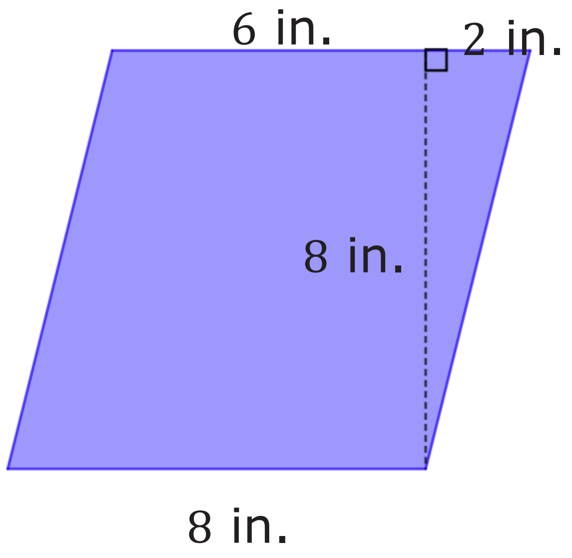
a. The store found 160 square yards (yd^2) of tile on sale. Based on the floor plan drawn, will the store have enough tiles to redo the flooring if they buy what is on sale?

b. Determine the amount of tile, in yd^2, the store is short or has left over.

8.3.5 Your Turn!

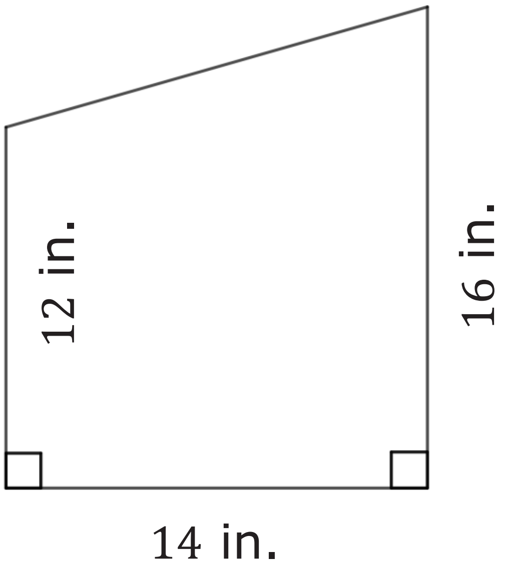
Find the area of each quadrilateral by decomposing. Lengths are shown in inches (in.). Include units.

1.



Area = ____________________

2.



Area = ____________________

3.

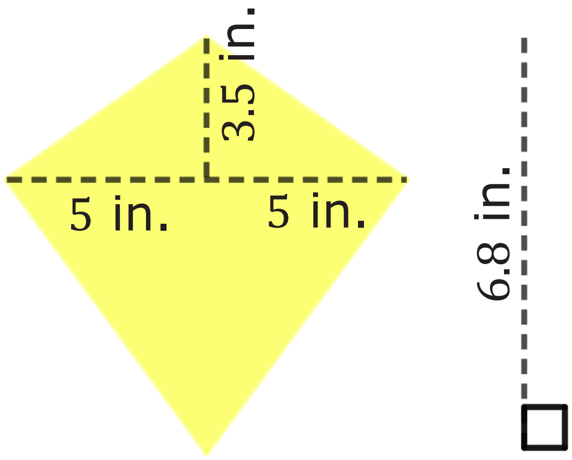
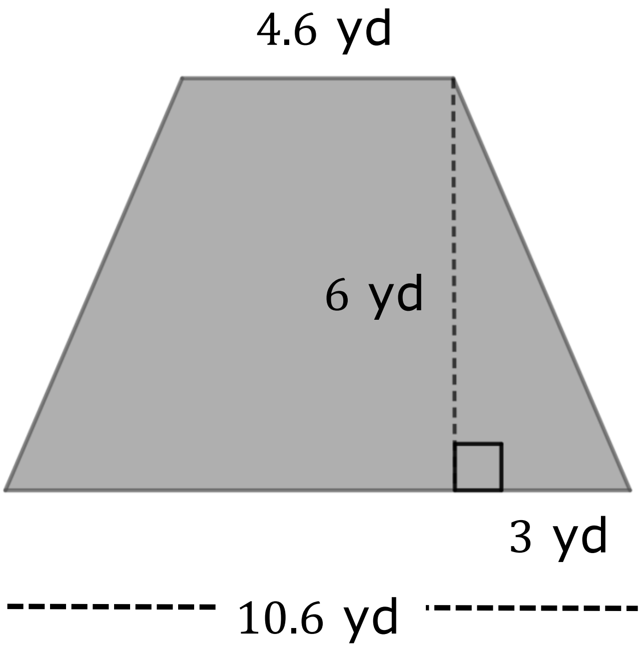


Image not drawn to scale

Area = ____________________

8.3.6 Wrap-Up: Ticket Out the Door

1. Find the area of the quadrilateral by decomposing. Lengths are given in yards (yd). Include units.



Area = ________________

